04. AI AGENTS

#Shiwani Jha

AI agent

- AI system that understands a goal → decides → uses tools → observe results → completes task.
- Core : Goal + Model + Tools + Content / Memory + Action Loop.

LLM v/s AI Agent

LLM	AI-agent
→ Generates response	→ Performs tasks / actions
→ Mainly single-step	→ can be multi-step
→ No tools inherently	→ use tools
→ User controls next step	→ Agent decides next action
→ Memory not inherent	→ can use memory.

Key difference: Action + tool use + decision-making.

- Real decision making
- Context samjhne ki ability
- Adaptive behaviour.

Core Components

- Goal / Instructions – what to achieve
- AI model – reasoning / decision-making
- Tools – external capabilities
- Memory – retained information
- Content – information available for current task.

Tools & Tool calling

- Tool : External function / service used by agent.
- Eg : API, Database, web search, calculator, maps, email.
- Tool calling : Model selects a tool + provides arguments → tool executes → result returned.
- Model decides ; tool performs.

Memory v/s Content

- Short-term memory : Curr. task / conservation information.
- Long – " " : stored information used later.
- Content : Information currently available to model
- Memory ≠ Context.

Agent-Loop

Goal → Understand → Plan → Choose Action → Tool/Action → Observe → Continue/Stop → Final Answer.

#) This repeated deide → act → observe cycle = Agent loop.

Planning

- Break complex goal into smaller steps.
- Eg: Find transport → hotel → check dates → compare → itinerary.

Decision Making

Agent decides : what action? , Need a tool? , which tool? What parameters? , Continue or finish?

Environment

External systems the agent interacts with : APIs, Databases, websites, Files, Applications, enterprise systems.

Agent Types

- Reactive : Input → Response
- Tool-using : Input → Tool → Result → Response
- Planning : Goal → Plan → Steps → Result
- Multi-agent : Multiple agents collaborate.

Assisted v/s Autonomous

- Assisted ⇒ Agent suggests → human approves ⇒ action.
- Autonomous ⇒ Agent decides → executes action.
- Level of autonomy on task + risk.

Agent v/s Chatbot

- Chatbot : Mainly communicates/responds.
- Agent : Communicates + decides + uses tools + performs actions + handles multi-step tasks.

Agent Failure Modes

→ wrong tool → wrong arguments → Hallucination
→ ∞ loop → Goal drift → Excessive tool usage
→ Instruction violation → Unsafe action

Guardrails

→ controls that keep agent with acceptable behaviour.
- No unauthorized data
- No dangerous commands

- No password exposure
- Approval for sensitive actions
- Stay within defined scope.

Human-in-the-loop

Human reviews/approves important actions.
Useful for: Emails, Purchases, Deleting data, Important record changes, Publishing content.

Agent Architecture

User → Agent → Model → Tools/memory/content → Action → Result → Observe → Continue/Finish.

Important Terms

- Agents : AI systems that can decide + act
- Tool : external capability
- Tool calling : Selecting/invoking tool
- Memory : Retained information
- Context : Current available info
- Planning : Breaking goal into steps
- Agent loop : Decides → Act → Observe.
- Environment : Ext. systems.
- Guardril : Behaviour Restrictions
- Human-in-loops : Human approval
- Multi-agent : Multiple collaborating agents.

AI agent ⇒ Goal + Model + Tools + Context/memory + Decision/ Action Loop
Agent loop ⇒ Plan → Act → Observe → Repeat
LLM generates, Agents can decide + act.
Main Failures : wrong tool + wrong arguments + Hallucination + loop + goal drift + Unsafe actions.

Top LLMs

GPT-5 : OpenAI
claude : Anthropic
Llama : Meta
Gemini Ultra - Google
Deepseek V3 - deepseek.